Solvency Field Theory V12.8

Primitive Anchor Origin:
Why $\{1,\ \frac{1}{2},\ \frac{1}{3}\}$ and Nothing Else

The capstone closing the operator-level
derivation chain from postulate to spectrum

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One founding postulate. One spacetime dimensionality. Three anchors. Nine modes. Zero free parameters.

Publication boundary. This document closes the operator-level derivation chain from V12.7 by tracing the three primitive support-domain anchors $\{1, 1/2, 1/3\}$ to native SFT structure. It does not provide a field-level PDE/action implementation. It does not derive physical masses, Ω_0, or a confirmed particle spectrum.

Contents

1 The physical picture

V12.7 derived two constraints — binary no-overwrite ($\rho \geq 1/2$) and 3+1 dimensional slot capacity ($N \leq 4$) — and showed that combining them with membership in a joint support domain selects exactly nine cadence modes. But V12.7 left one question open: why are the three primitive anchors of the support domain $\{1, 1/2, 1/3\}$ and not some other set?

V12.8 answers this. The three anchors are not chosen. They are forced.

1.1 Three anchors from three structures

Every maintained worldline in SFT has three structural features that are not optional:

It ticks. It cycles at its Compton frequency, maintaining itself tick by tick. The identity cadence — the fact that a worldline has a base period at all — is the most basic feature of a maintained structure. The cadence ratio of the base cycle to itself is $\rho = 1$.

It writes. Each tick inscribes a Planck area on the boundary. Each write must be distinguishable from the previous write under no-overwrite. The minimum angular footprint for distinguishable binary writes is half a turn (from V12.7's binary no-overwrite chain). The cadence ratio of the binary write cell is $\rho = 1/2$.

It has three transverse directions. A timelike worldline in 3+1 spacetime has three spatial directions perpendicular to it. The transverse frame distributes across these three directions. The simplest equal distribution assigns 1/3 of the frame rotation to each direction. The cadence ratio of the tri-normal virtual support is $\rho = 1/3$.

These are not three independent assumptions. They are three aspects of one object — the maintained worldline tube in 3+1 spacetime. The tube ticks (giving 1), writes binary records (giving 1/2), and has three transverse directions (giving 1/3). There is nothing else to add and nothing that can be removed.

1.2 Uniqueness

R100 tested every alternative and found that $\{1, 1/2, 1/3\}$ is the unique minimal anchor set that generates the exact nine-mode seed.

Omitting any anchor breaks the seed: without 1, the identity mode is lost; without 1/2, the binary branch collapses; without 1/3, the mode at 5/3 is lost.

Replacing 1/3 with 1/4 (four transverse directions, as in 4+1 spacetime) generates the wrong seed: it loses 5/3 and admits the intruder 7/4.

Adding 1/4 as a fourth anchor over-expands the support domain and admits 7/4.

No other dimensionality works. The tri-normal scan tested $D_\perp$ from 1 through 6. Only $D_\perp = 3$ produces the exact seed. The binary scan tested K from 1 through 6. Only $K = 2$ produces the exact seed.

The three anchors are locked by the intersection of two uniqueness results: only the binary write cell ($K = 2$) and only three transverse dimensions ($D_\perp = 3$) produce the correct seed when combined with the identity cadence.

1.3 The closed chain

With V12.8, the operator-level derivation chain from the founding postulate to the nine-mode spectrum is complete:

1. **Founding postulate:** existence is active maintenance; each tick writes one Planck area; writes cannot be overwritten.
2. **Binary predicate** (R94): no-overwrite asks same/different, requiring exactly $K = 2$ distinguishable states.
3. **Half-turn cell** (R92–R93): $K = 2$ on a compact circle gives cell width 1/2, the angular Planck bit.
4. **Non-overlap update** (R95): consecutive binary cells must not overlap, giving $\rho \geq 1/2$.
5. **Dimensional capacity** (R96): 3+1 spacetime provides $1+3 = 4$ independent phase channels, giving $N \leq 4$.
6. **Identity anchor** (R100): the base cadence $\rho = 1$ is the unique joint fixed point of the reflection and reciprocal maps.
7. **Binary anchor** (R100): the write-cell cadence $\rho = 1/2$ is the unique binary anchor ($K = 2$) that preserves the seed.
8. **Tri-normal anchor** (R100): the normal-share cadence $\rho = 1/3$ is the unique $D_\perp = 3$ anchor that preserves the seed.
9. **Support domain:** joint closure of $\{1, 1/2, 1/3\}$ under $R(\rho) = 2 - \rho$ and $C(\rho) = 1/\rho$ generates the support domain D_{RC}.
10. **Nine-mode seed:** $S = \{\rho \in D_{RC} : \rho \geq 1/2,\ N \leq 4\} = \{1/2, 2/3, 3/4, 1, 5/4, 4/3, 3/2, 5/3, 2\}$.

Every step traces to: one founding postulate (no-overwrite) plus one geometric input (3+1 dimensionality). Nothing else enters.

2 The R100 audit

2.1 Anchor origin scorecard

R100 tested twelve anchor configurations. The results:

Configuration	**Modes**	**Targets**	**Status**
No anchors	0	0/5	Fail
Identity only	1	1/5	Fail
Binary only	7	4/5	Fail: missing 1, 5/3
Tri-normal only	1	0/5	Fail
Identity + binary	8	5/5	Fail: missing 5/3
Identity + tri-normal	2	1/5	Fail
Binary + tri-normal	8	4/5	Fail: missing 1
Identity + binary + tri-normal	**9**	**5/5**	**Exact seed**
Replace 1/3 with 1/4	9	5/5	Fail: wrong seed
Add 1/4	10	5/5	Fail: 7/4 intruder
All rationals	10	5/5	Fail: 7/4 intruder
No support filter	10	5/5	Fail: 7/4 intruder

Only $\{1, 1/2, 1/3\}$ produces the exact nine-mode seed. Every weakening, strengthening, or substitution fails.

2.2 Tri-normal dimension scan

$D_\perp$	Anchor	Modes	Exact?	Issue
1	1	8	No	Missing 5/3
2	1/2	8	No	Missing 5/3
3	**1/3**	**9**	**Yes**	**Exact seed**
4	1/4	9	No	Admits 7/4, loses 5/3
5	1/5	8	No	Missing 5/3
6	1/6	8	No	Missing 5/3

Only $D_\perp = 3$ — three transverse directions, as in 3+1 spacetime — produces the exact seed.

2.3 Binary K scan

K	Anchor	Modes	Exact?	Issue
1	1	2	No	Signal lost
2	**1/2**	**9**	**Yes**	**Exact seed**
3	1/3	2	No	Signal lost
4	1/4	3	No	Admits 7/4
5	1/5	2	No	Signal lost
6	1/6	2	No	Signal lost

Only $K = 2$ — the binary minimum nontrivial distinction — produces the exact seed.

3 What is earned, conditional, and open

3.1 Earned

The primitive anchor set $\{1, 1/2, 1/3\}$ traces to identity cadence, binary no-overwrite, and three transverse dimensions.

It is the unique minimal tested anchor set that generates the exact seed under V12.7 filters.

$D_\perp = 3$ is the unique dimensionality and $K = 2$ is the unique binary choice that produce the exact seed.

Omitting any anchor breaks the seed. Substituting any anchor either loses modes or admits intruders.

The operator-level derivation chain from the founding postulate to the nine-mode spectrum is closed.

3.2 Open

Field-level PDE/action derivation of the anchors and maps.

Mapping from nine toy cadence modes to physical particle species.

Mass hierarchy and Standard Model masses.

Absolute cadence scale Ω_0.

Envelope normalization and coupling constants.

4 The complete derivation chain

For reference, the full operator-level chain from postulate to spectrum, assembled from V12.4 through V12.8:

Step	Statement	Source	Status
1	Existence is active maintenance; writes cannot be overwritten.	Founding postulate	Axiom
2	No-overwrite is a binary predicate.	R94A/B	Pass
3	K=2 is the minimum faithful record.	R94A/B	Pass
4	Half-turn cell: width $= 1/K =$ $1/2$.	R92–R93	Pass
5	Non-overlap update: $\rho \geq 1/2$.	R95	Pass
6	3+1 slot capacity: $N \leq 4$.	R96	Pass
7	Identity anchor: $\rho = 1$.	R100	Pass
8	Binary anchor: $\rho = 1/2$.	R100	Pass
9	Tri-normal anchor: $\rho = 1/3$.	R100	Pass
10	Joint R/C support domain.	R98–R99	Pass
11	Nine-mode seed.	R88 + chain	Pass

Inputs: one postulate, one dimensionality.
Output: nine modes.
Free parameters: zero.

5 The nine modes

$$\boxed{\tfrac{1}{2},\quad \tfrac{2}{3},\quad \tfrac{3}{4},\quad 1,\quad \tfrac{5}{4},\quad \tfrac{4}{3},\quad \tfrac{3}{2},\quad \tfrac{5}{3},\quad 2} \tag{1}$$

These are the complete set of cadence ratios ρ satisfying:

1. $\rho \in D_{RC} = \mathrm{closure}_{R,C}(\{1, 1/2, 1/3\})$,
2. $\rho \geq 1/2$,
3. $\mathrm{denom}(\rho) \leq 4$.

No tenth mode satisfies all three. No eight-element subset is closed under the support maps. The nine-mode seed is sharp: every weakening admits intruders, every strengthening kills valid modes.

Whether these nine abstract cadence ratios correspond to nine real particle species — and if so, which ones — is the frontier beyond V12.8.

6 Conclusion

V12.8 closes the operator-level chain. The founding postulate (no-overwrite) plus the dimensionality of spacetime (3+1) determine three primitive anchors (1, 1/2, 1/3), two constraints ($\rho \geq 1/2$, $N \leq 4$), one support domain, and one nine-element spectrum. Nothing is free. Nothing is fitted. The spectrum is what you get — the only thing you can get — when you ask which winding patterns can maintain themselves under binary no-overwrite distinguishability in three spatial dimensions plus time.

The particles that exist are the ones that can answer a yes/no question on a compact phase circle in 3+1 dimensions. There are nine of them.

The anchors that seed them are the identity cadence, the binary write cell, and the three transverse directions of spacetime. There is nothing else to add.

Acknowledgments

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References

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